

Summary of Bug fixes and Enhancements included in BSIM-IMG 103.0.0

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A. Summary of bug fixes done in BSIM-IMG 103.0.0

1. 2018bug3 (Mentor graphics): Vthop is independent of PHIG1
2. 2018bug4 (ADI): Division by zero in vdsx term of DITS
3. 2018bug5 (ADI): Checks for DSUB_i < 0, DVT1_i < 0
4. 2018bug6 (ADI): UDB2_i needs correction. Redundant variable removed.
5. 2018bug7 (ADI): Remove Redundant Lines
6. 2018bug8 (ADI): gcrq term initialization in else block
7. 2018bug10 (Infineon/ADI): Implementation of \$simparam
8. 2019bug1 (ADI): Apply Vds swapping in GDS OP
9. 2019bug3 (ADI): Use of OPM/OPD macros in OP Points
10. 2019bug4: Redundant definition of V(s, si) and V(d, di) when RDSMOD==2

B. Summary of enhancements done in BSIM-IMG 103.0.0

- 2018enh4: Set XL as model parameter
- 2019enh1 (GF): 1/f noise model needs improvement in subthreshold region
- 2019enh3 (Cadence): Add ROUT OP. variable

C. Description of bug fixes done in BSIM-IMG 103.0.0

1. 2018bug3 (Mentor graphics): Vthop is independent of PHIG1

Issue: Vth equation implemented inside the OP section is independent of PHIG1.

BSIM-IMG 103.0.0beta5

```
Vthop = devsign * (T1 + dvth_all + DELVTRAND);
```

BSIM-IMG 103.0.0

```
Vthop = -devsign * (PHIG1_i - PhiRef + T1 + dvth_all + DELVTRAND);
```

2. 2018bug4 (ADI): Division by zero in vdsx term of DITS

BSIM-IMG 103.0.0beta5

```
dvth_dibl = -(ETA0_t + ETAB_i * vbgs) * Theta_DIBL * (vdsx + ETA1_i * sqrt(vdsx + 0.01))  
+ (DVTP0_i * Theta_DITS * pow(vdsx, DVTP1_i));
```

BSIM-IMG 103.0.0

```
dvth_dibl = -(ETA0_t + ETAB_i * vbgs) * Theta_DIBL * (vdsx + ETA1_i * sqrt(vdsx + 0.01))  
+ (DVTP0_i * Theta_DITS * pow((vdsx + 0.01), DVTP1_i));
```

3. 2018bug5 (ADI): Checks for DSUB_i < 0, DVT1_i < 0

Checks are added for DVT1_i and DSUB_i in [BSIM-IMG 103.0.0beta6](#)

```
if (DVT1_i <= 0.0) begin  
    $strobe("Fatal: DVT1_i = %e is not positive.", DVT1_i);  
    $finish(0);  
end  
  
if (DSUB_i <= 0.0) begin  
    $strobe("Fatal: DSUB_i = %e is not positive.", DSUB_i);  
    $finish(0);  
end
```

4. 2018bug6 (ADI): UDB2_i is not set prior to this line

BSIM-IMG 103.0.0beta5

```
UDB2_i = UDB2_i + AADB2 * lexp(-Leff / BUDB2);
```

BSIM-IMG 103.0.0

```
UDB2_i = UDB2 + AADB2 * lexp(-Leff / BUDB2);
```

Redundant variable removed

BSIM-IMG 103.0.0beta5

```
real eta_mu, eta_mu_cv, Eeffm, Eeffm2, Eeffm_cv, Dmob, Dtot, Dmob_cv, u0, utotal;
```

BSIM-IMG 103.0.0

```
real eta mu, eta mu cv, Eeffm, Eeffm2, Eeffm cv, Dmob, Dmob cv, utotal;
```

5. 2018bug7 (ADI): Remove Redundant Lines

BSIM-IMG 103.0.0beta5

```
xg1 = vgfb1eff / nVtm;  
xg2 = (vgfb2 - dvth_all + DELVTRAND) / nVtm;  
//////////BEGIN CORE CALCULATIONS  
//////////saturation potential at back  
PHISSatback = ln((k1_2*(xg1-phi1_0)*(xg1-phi1_0)+  
PHISSat = ln((k1_2*(xg1-phi1_0)*(xg1-phi1_0)+  
T3 = (phi1_0*(1.0 + k1)-phi2)/k1;  
T4 = ln((k1_2*(T3-phi1_0)*(T3-phi1_0)+39.4784  
T5 = T4-phi1_0;  
//PHISSat = PHISSat-T5;  
PHISSatback2 = (PHISSat + k2 * xg2) / (1.0 +  
//phi2 guess using only PHISSatback and PHISSat  
phi2sub= xg2 + keq_k2* (xg1 - xg2);  
  
phi2 = min(phi2sub,PHISSatback);
```

BSIM-IMG 103.0.0

```
xg1 = vgfb1eff / nVtm;  
xg2 = (vgfb2 - dvth_all + DELVTRAND) / nVtm;  
//////////BEGIN CORE CALCULATIONS  
//////////saturation potential at back  
PHISSatback = ln((k1_2*(xg1-phi1_0)*(xg1-phi1_0)+  
PHISSat = ln((k1_2*(xg1-phi1_0)*(xg1-phi1_0)+  
  
PHISSatback2 = (PHISSat + k2 * xg2) / (1.0 +  
//phi2 guess using only PHISSatback and PHISSat  
phi2sub= xg2 + keq_k2* (xg1 - xg2);  
  
phi2 = min(phi2sub,PHISSatback);
```

6. 2018bug8 (ADI): gcrq term initialization in else statement

BSIM-IMG 103.0.0beta5

```
// NQS gate resistance Ref: BSIM4  
if (NQS_MOD==1 && XRCRG1_i!=0.0) begin  
    T0 = utotal * cox1 * Weff / Leff;  
    IdsovVds = T0 * qia;  
    gcrq = NF * XRCRG1_i * (IdsovVds +  
end
```

BSIM-IMG 103.0.0

```
if (NQSMOD == 1 && XRCRG1_i != 0.0) begin
    T0 = utotal * cox1 * Weff / Leff;
    IdsovVds = T0 * qia;
    gcrg = NF * XRCRG1_i * (IdsovVds + XRCF
end else begin
    gcrg = 0;
end
```

7. 2018bug10 (Infineon/ADI): Implementation of \$simpam

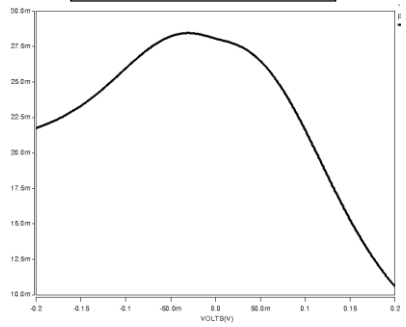
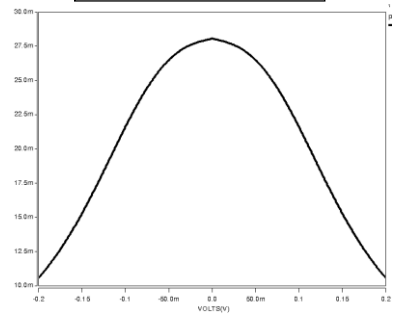
BSIM-IMG 103.0.0beta5

```
if (sigvds > 0.0) begin
    I(di, si) <+ devsign * ids + (GDSMIN * V(di, si));
    I(di, si) <+ devsign * (igidl + Iii);
    I(si, di) <+ devsign * igisl;
    I(gi, si) <+ devsign * (igcs + igs);
    I(gi, di) <+ devsign * (igcd + igd);
end else begin
    I(si, di) <+ devsign * ids + (GDSMIN * V(si, di));
    I(si, di) <+ devsign * (igidl + Iii);
    I(di, si) <+ devsign * igisl;
    I(gi, di) <+ devsign * (igcs + igs);
    I(gi, si) <+ devsign * (igcd + igd);
end
```

BSIM-IMG 103.0.0

```
if (sigvds > 0.0) begin
    I(di, si) <+ devsign * ids + ($simpam("gmin",1e-12) * V(di, si));
    I(di, si) <+ devsign * (igidl + Iii);
    I(si, di) <+ devsign * igisl;
    I(gi, si) <+ devsign * (igcs + igs);
    I(gi, di) <+ devsign * (igcd + igd);
end else begin
    I(si, di) <+ devsign * ids + ($simpam("gmin",1e-12) * V(si, di));
    I(si, di) <+ devsign * (igidl + Iii);
    I(di, si) <+ devsign * igisl;
    I(gi, di) <+ devsign * (igcs + igs);
    I(gi, si) <+ devsign * (igcd + igd);
end
```

8. 2019bug12 (ADI): Apply Vds swapping in GDS OP

BSIM-IMG 103beta5	BSIM-IMG 103.0.0
<pre>// Conductances IDS = devsign * sigvds * ids; GM = ddx(IDS, V(gi)); GMBS = ddx(IDS, V(bg)); GDS = ddx(IDS, V(di));</pre>	<pre>if (sigvds > 0.0) begin VFgs = V(fg, s); VFgd = V(fg, d); VDS = V(d, s); VSBg = V(s, bg); VDBg = V(d, bg); GDS = ddx(IDS, V(di)); ROUT = 1.0/GDS; end else begin VFgs = V(fg, d); VFgd = V(fg, s); VDS = V(s, d); VSBg = V(d, bg); VDBg = V(s, bg); GDS = ddx(-IDS, V(si)); ROUT = 1.0/GDS; end</pre>
Asymmetric GDS 	Symmetric GDS 

9. 2019bug3 (ADI): Use of OPM/OPD macros in OP Points

The operating-point variables in BSIM-IMG 103.0.0beta5 are not using the new OPM and OPD macros. These new macros are implemented in BSIM-IMG 103.0.0beta6 similar to BSIM-BULK beta code.

Charges, currents, capacitances, and conductance's should use OPM.
Resistances should use OPD.
Voltages continue to use OPP.

10. 2019bug4: Redundant definition of V(s, si) and V(d, di) when RDSMOD==2

BSIM-IMG has the contributions to (d,di) and (s,si) done in two different blocks. We

have combined these two blocks to eliminate the internal nodes.

BSIM-IMG 103.0.0beta5

```
if (RDSMOD == 2) begin
    V(d, di) <+ 0.0;
    V(s, si) <+ 0.0;
end else begin
    gdpr = 1.0 / Rdrain;
    gspr = 1.0 / Rsource;
    I(d, di) <+ white_noise(Nt * gdpr, "Rd");
    I(s, si) <+ white_noise(Nt * gspr, "Rs");
end

if (RDSMOD == 2) begin
    V(d, di) <+ 0.0;
    V(s, si) <+ 0.0;
end else begin
    I(d, di) <+ V(d, di) / Rdrain;
    I(s, si) <+ V(s, si) / Rsource;
end
```

BSIM-IMG 103.0.0

```
if (RDSMOD == 2) begin
    V(d, di) <+ 0.0;
    V(s, si) <+ 0.0;
end else begin
    gdpr = 1.0 / Rdrain;
    gspr = 1.0 / Rsource;
    I(d, di) <+ V(d, di) * gdpr;
    I(s, si) <+ V(s, si) * gspr;
    I(d, di) <+ white_noise(Nt * gdpr, "Rd");
    I(s, si) <+ white_noise(Nt * gspr, "Rs");
end
```

D. Description of enhancements done in BSIM-IMG 103.0.0

1. 2018enh4: Set XL as model parameter

BSIM-IMG 103.0.0beta5

```
// Variability Handles
`IPRnb ( XL , 0.0
```

BSIM-IMG 103.0.0

```
`MPRnb ( XL , 0.0
```

2. 2019enh1 (GF): 1/f noise model needs improvement in subthreshold region

BSIM-IMG 103.0.0beta5

```
T1 = `q * `q * `q * Vtm * abs(ids) * utotal;
T2 = 1.0e10 * coxeff * Leffnoisq;
N0 = coxeff * qis / `q;
N1 = coxeff * qid / `q;

Nstar = Vtm / `q * (coxeff + CIT_i);
T3 = NOIA * lln((N0 + Nstar) / (N1 + Nstar));
T4 = NOIB * (N0 - N1);
T5 = 0.5 * NOIC * (N0 * N0 - N1 * N1);
T6 = `q * Vtm * ids * ids;

T7 = 1.0e10 * Leffnoisq * Weff * NF;
T8 = NOIA + NOIB * N1 + NOIC * N1 * N1;
T9 = (N1 + Nstar) * (N1 + Nstar);
Ssi = T1 / T2 * (T3 + T4 + T5) + T6 / T7 * DelC1m * T8 / T9;

T10 = NOIA * `q * Vtm;
T11 = Weff * NF * Leffnoi * 1.0e10 * Nstar * Nstar;
Swi = T10 / T11 * ids * ids;
```

BSIM-IMG 103.0.0

```
if (FNM0D == 1) begin
    T1 = qia2/QSREF_i;
    T2 = 1 + pow(T1, MPOWER_i);
    T3 = NOIA2_i/T2;
    T4 = T3 / NOIA;
    T5 = 0.5 * (T4 + 1 + sqrt((T4 - 1) * (T4 - 1) + 0.25 * SMOOTH * SMOOTH));
    NOIAeff = NOIA * T5;
end else begin
    NOIAeff = NOIA;
end

T1 = `q * `q * `q * Vtm * abs(ids) * utotal;
T2 = 1.0e10 * coxeff * Leffnoisq;
N0 = coxeff * qis / `q;
N1 = coxeff * qid / `q;

Nstar = Vtm / `q * (coxeff + CIT_i);
T3 = NOIAeff * lln((N0 + Nstar) / (N1 + Nstar));
T4 = NOIB * (N0 - N1);
T5 = 0.5 * NOIC * (N0 * N0 - N1 * N1);
T6 = `q * Vtm * ids * ids;

T7 = 1.0e10 * Leffnoisq * Weff * NF;
T8 = NOIAeff + NOIB * N1 + NOIC * N1 * N1;
T9 = (N1 + Nstar) * (N1 + Nstar);
Ssi = T1 / T2 * (T3 + T4 + T5) + T6 / T7 * DelC1m * T8 / T9;

T10 = NOIAeff * `q * Vtm;
T11 = Weff * NF * Leffnoi * 1.0e10 * Nstar * Nstar;
Swi = T10 / T11 * ids * ids;
```

3. 2019enh3 (Cadence): Add ROUT OP. variable

BSIM-IMG 103beta5	BSIM-IMG 103.0.0
ROUT op. was not present in this version.	<pre>if (sigvds > 0.0) begin VFGS = V(fg, s); VFGD = V(fg, d); VDS = V(d, s); VSBG = V(s, bg); VDBG = V(d, bg); GDS = ddx(IDS, V(di)); ROUT = 1.0/GDS; end else begin VFGS = V(fg, d); VFGD = V(fg, s); VDS = V(s, d); VSBG = V(d, bg); VDBG = V(s, bg); GDS = ddx(-IDS, V(si)); ROUT = 1.0/GDS; end</pre>
	